

# Rising Beyond Borders

## Document 25 — Production Deployment, Launch & Operations

### Master Specification + Claude Code Implementation Prompt

**Status:** Next phase after STAGING-READY. This document defines production deployment and launch operations; it does not authorize unapproved RBB content, credentials, providers, tracking, donations, or forms.

## 1. Purpose

Move the approved staging implementation toward a controlled production launch with clear environment separation, secure configuration, repeatable deployment, smoke testing, rollback, and post-launch operations. This is an infrastructure/release phase, not a redesign.

Document 24 remains the source of truth for production readiness blockers and approvals. Do not invent missing RBB content or silently activate gated services.

## 2. Current Starting State

- **Staging:** complete and visually populated with temporary demo content.
- **Production:** demo content excluded; release gate exists; production remains blocked until approvals and infrastructure/provider setup are complete.
- **Forms:** implementation exists; production activation remains approval/configuration dependent.
- **Donations:** Razorpay architecture exists; credentials and final donation decisions are required.
- **Email:** Resend integration exists; production domain, sender, recipients and rotated credentials are required.
- **Analytics/observability:** disabled by default and require explicit approval.
- **Images:** staging demo assets are isolated; production asset licensing/approval remains part of release readiness.

## 3. Environment Model

| Environment | Purpose          | Demo | Real email    | Payments      | Indexing             |
|-------------|------------------|------|---------------|---------------|----------------------|
| Development | Developer work   | No   | No            | No            | No                   |
| Staging     | Full demo/review | Yes  | Test only     | Test/stand-in | No                   |
| Production  | Public RBB site  | No   | Approved only | Approved only | Approved routes only |

## 4. Production Infrastructure

- Record the actual RBB-approved production domain and hosting platform; never guess them.
- Enforce HTTPS and redirect HTTP to HTTPS.
- Apply host-level security headers, including HSTS only after HTTPS is confirmed.
- Implement real permanent HTTP 301/308 redirects at the host rather than relying only on client-side forwarding.
- Unknown routes must return a real HTTP 404 while valid application routes continue to work.
- Document DNS, TLS, fallback routing, cache/CDN behavior and deployment ownership.

## 5. Environment Variables & Secrets

- Keep all secrets server-side. Never place them in Vite client variables, source, HTML, build metadata, logs or commits.
- Resend: EMAIL\_API\_KEY / RESEND\_API\_KEY as supported by the existing server configuration.
- Razorpay: RAZORPAY\_KEY\_ID, RAZORPAY\_KEY\_SECRET, RAZORPAY\_WEBHOOK\_SECRET.
- Production mail: EMAIL\_FROM plus approved recipients.
- Donation settings: DONATION\_ENABLED, currency and approved amount constraints.
- Forms: EMAIL\_MODE=live only after approval and provider verification.
- VITE\_SITE\_URL must be the actual HTTPS production origin.

**Security:** the previously shared Resend API key must be revoked/rotated and must not be reused.

## 6. Resend Production Activation

- Verify the RBB-owned sending domain.
- Configure the approved From address and recipients.
- Confirm acknowledgement behavior and wording.
- Run test-mode delivery before live mode.
- Verify provider/domain authentication and delivery.
- Confirm logs expose neither API keys nor unnecessary personal submission data.
- Switch EMAIL\_MODE to live only after all approvals and checks pass.

## 7. Razorpay Production Activation

- Confirm the RBB Razorpay production account and credentials.
- Confirm currency, allowed amounts, custom amount policy, donor fields and designation.
- Configure and verify the webhook endpoint and secret.
- Verify order/signature checks server-side.
- Treat captured/webhook state as authoritative; browser callbacks are not authoritative.
- Test retries, duplicate callbacks and idempotency.
- Confirm payment/refund email behavior.
- Use exact Razorpay browser origins in CSP; never use a wildcard.

## 8. Forms & Anti-Spam

- Keep public forms disabled until Privacy Policy and operational recipients are approved.
- Verify same-origin /api/forms routing.
- Verify server-side validation, size limits and unknown-field rejection.
- Verify honeypot/minimum-fill-time controls and generic public errors.
- Replace in-memory rate limiting with host/shared backing if production is multi-instance.
- Verify duplicate-submission handling.
- Keep logs privacy-minimal and reference-based.

## 9. Privacy, Cookies & Third-Party Technology

- Publish the approved Privacy Policy before collecting form/newsletter data.
- Publish Terms and Accessibility information when approved.

- Decide whether a Cookies page is required after the final production technology audit.
- Audit the final host for cookies, analytics, monitoring, support widgets and injected scripts.
- Review existing font origins against the approved privacy wording.
- Do not enable analytics, Search Console, session replay or monitoring SDKs without explicit approval.

## 10. SEO & Indexing

- Use the real HTTPS production origin for canonical URLs.
- Sitemap contains only approved indexable routes.
- Production robots.txt permits approved indexing.
- Staging remains blocked/noindex.
- Unknown/draft/unapproved content remains excluded.
- Host redirects are permanent.
- Validate social metadata/images when approved assets exist.

## 11. Build & Release Artifact Gate

- Run content validation and the production release gate.
- Run security/secret scans, accessibility tests and server tests.
- Run hydration/browser smoke tests.
- Verify zero demo images/text/files in the production artifact.
- Verify no .env files, source maps, fixtures, staging routes or development catalogue content ship.
- Record a non-sensitive build revision for rollback identification.

## 12. Production Smoke Test

- Homepage and approved routes load over HTTPS.
- Desktop and mobile navigation work.
- Unknown route returns true 404.
- All permanent redirects resolve correctly.
- Images load without failures.
- Keyboard focus and skip link work.
- Forms behave according to their configured activation state.
- Resend delivery is confirmed if forms are live.
- Razorpay flow/webhook is confirmed according to the approved launch procedure if donations are live.
- No console errors or CSP violations.
- No unexpected cookies/storage/third-party requests.
- Robots, sitemap and canonicals are correct.
- No demo content is discoverable.

## 13. Rollback

- Keep the previous known-good production build identifiable.
- Make deployment reversible without manual content editing.

- Restore the previous artifact/build if a critical release defect occurs.
- Disable newly activated forms or donations through server-side configuration when appropriate.
- Recheck DNS, redirects, CSP and environment settings after rollback.
- Record the rollback reason and affected build revision.

## 14. Post-Launch Monitoring

- Monitor uptime and HTTP errors when an approved provider is enabled.
- Monitor form delivery failures and provider responses.
- Monitor Razorpay webhook/payment failures if donations are live.
- Monitor CSP violations.
- Monitor approved performance metrics.
- Watch for 404/redirect spikes.
- Audit unexpected third-party requests after host deployment.
- Do not collect unnecessary personal data in logs.

## 15. Launch-Day Sequence

1. Confirm all Document 24 blockers are resolved or explicitly accepted by RBB.
2. Freeze approved production content.
3. Confirm domain, DNS and HTTPS.
4. Confirm host security headers and permanent redirects.
5. Configure production variables/secrets.
6. Verify Resend domain, sender and recipients.
7. Verify Razorpay configuration if donations launch.
8. Build the release candidate and run every automated gate.
9. Deploy the exact release artifact.
10. Run production smoke tests.
11. Verify robots, sitemap and canonicals.
12. Verify forms/payment flows according to activation state.
13. Record release revision and deployment time.
14. Begin the agreed post-launch monitoring window.

## 16. Do Not Launch Until

- Production domain and host are selected.
- HTTPS and host-level security configuration are verified.
- Required RBB content and policies are approved/published.
- Image ownership/licensing decisions are resolved.
- Resend credentials are rotated and sender/recipients verified.
- Razorpay configuration is verified if donations are enabled.
- Production rate limiting is safe for the host architecture.
- Release gate passes.

- Production artifact contains no demo content.
- Production smoke test passes.

## 17. Operational Documentation

- ``docs/DEPLOYMENT.md`` — deployment/host requirements.
- ``docs/SECURITY.md`` — security boundaries and headers.
- ``docs/FORMS_AND_EMAIL.md`` — form/email activation.
- ``docs/DONATIONS.md`` — Razorpay activation and states.
- ``docs/RELEASE_READINESS.md`` — blockers and approvals.
- ``docs/IMAGE_INVENTORY.md`` — asset source/licensing.
- ``docs/DEMO_CONTENT.md`` and ``docs/DEMO_IMAGE_USAGE.md`` — staging-only material.

## 18. Maintenance Boundary

Routine content changes continue through the approved content workflow. Infrastructure changes are reviewed separately. Do not introduce a CMS, admin dashboard, database, analytics provider, payment provider or third-party widget as a convenience without a new architecture and approval decision.

## 19. Claude Code Implementation Prompt

You are implementing Document 25 — Production Deployment, Launch & Operations for Rising Beyond Borders.

Treat Documents 1–24 and the current repository as authoritative. Do not redesign the site and do not invent RBB content.

Objective:

Prepare a controlled production deployment and operational release process while preserving strict separation between development, staging/demo and production.

Requirements:

1. Audit deployment, environment, build, release-gate, security, forms/email and donation implementation.
2. Preserve ``npm run dev`` as production-safe and ``npm run dev:staging`` as demo/staging.
3. Ensure production builds contain zero demo content, demo images, development catalogue content, staging routes, test fixtures, secrets or source maps.
4. Preserve the validated production-domain mechanism.
5. Document host requirements for HTTPS, permanent redirects, true 404 responses, routing fallback and security headers.
6. Preserve strict CSP; add only exact Razorpay origins required when donations are enabled. Never add wildcards or `unsafe-eval`.
7. Keep Resend server-side only. Never hard-code credentials. Require a rotated production key.
8. Keep Razorpay secrets server-side and preserve server-side order creation, signature verification, webhook verification, idempotency and authoritative states.
9. Do not enable production forms or donations unless configuration and approval gates are satisfied.
10. Ensure production rate limiting is appropriate for the hosting architecture; flag in-memory rate limiting if multi-instance deployment is possible.
11. Keep analytics/Search Console/monitoring disabled unless explicitly approved and configured.
12. Update deployment, launch, smoke-test and rollback documentation.
13. Run all existing tests and add only focused deployment/release checks where necessary.
14. Do not stage or commit files.

Final report must include:

- files changed
- deployment/environment assumptions
- commands run
- test results
- production artifact checks
- demo-content leak result
- secret scan result
- CSP/security result
- accessibility result

- forms result
- donation result
- unresolved RBB decisions/blockers
- exact recommended next action

Do not claim production-ready unless the release gate and actual deployment configuration support that conclusion.

## 20. Definition of Done

- A documented production deployment path exists.
- Environment separation remains intact.
- Production build is deterministic and demo-free.
- Secrets remain server-only and are not committed.
- Host HTTPS, redirects, headers and 404 requirements are documented.
- Resend/Razorpay activation paths are controlled and testable.
- Forms/donations cannot accidentally activate without configuration/approval.
- Rollback is documented.
- Production smoke tests are repeatable.
- The final report distinguishes technical completion from RBB approval.

## 21. RBB Decisions Required

- Production domain and hosting platform.
- DNS/TLS ownership and configuration.
- Resend sending domain, From address and recipients.
- Razorpay production account, currency, amount rules and payment policy.
- Form recipients and acknowledgement wording.
- Privacy/Terms/Accessibility/Cookies approval.
- Analytics/observability decisions.
- Image licensing/ownership approval.
- Final content and SEO approval.
- Post-launch monitoring and incident contact.

## 22. Release Principle

**Technical readiness is not the same as RBB approval.** A feature can be implemented and tested while remaining disabled in production. The release process must preserve that distinction.